

NIKITA TULI

Chicago, IL

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Computational Genomic Medicine graduate student with a background in computer science, machine learning, and healthcare-focused data analytics. Interested in computational biology, bioinformatics, AI for biomedical research, and data-driven clinical applications.

Education

Case Western Reserve University – School of Medicine

M.S. in Computational Genomic Medicine

Expected Graduation: Spring 2028

University of Minnesota – Twin Cities

BS in Computer Science

August 2023 - May 2026

Skills

Programming: Python, SQL, Java, C++, JavaScript, R, Linux, Bash

Data Science & ML: Data Analysis, Statistics, Machine Learning, Predictive Modelling, Convolutional Neural Networks (CNNs)

Tools: Git/GitHub, Power BI, Seeq, Pi Vision, Excel, Power Apps, Copilot Studio, Pandas, NumPy, Jupyter, Bioconductor, BLAST

Software Development: HTML/CSS/JavaScript, Object-Oriented Programming, Data Structures and Algorithms

Domain Interests: Computational Biology, Genomics, Biomedical Data Science, Healthcare AI

Experience

Shirley Ryan Ability Lab Hospital

Data Analytics/Patient Services Intern

May 2026 – Present

- Designed and implemented AI-driven tools to automate administrative and operational workflows in a clinical setting, improving efficiency and allowing staff to focus more on patient care.
- Created applications that saved an estimated amount \$50,000 a year.
- Partnered with clinical, operational, and technical stakeholders to identify workflow bottlenecks, define system requirements, and support deployment of data-informed solutions.
- Managed end-to-end implementation of internal AI tools, including planning, testing, validation, and iterative refinement based on user feedback.
- Created presentations and project updates for leadership, translating technical progress and operational impact for non-technical audiences.
- Participated in Grand Rounds and Research Rounds, gaining exposure to clinical workflows, rehabilitation research, and healthcare innovation.

Solvay Chemicals

Digital Transformations Intern

May 2025 – April 2026

- Developed custom digital tools and Power Apps that replaced manual workflows and improved operational efficiency across the Green River facility.
- Built and trained CNNs using machinery sensor and operational data to support predictive maintenance and reduce unplanned downtime.
- Created dashboards and internal reporting tools in Power BI, Seeq, and PI Vision to monitor performance metrics and downtime trends in real time.
- Transformed, and integrated large operational datasets using Python, SQL, Java, and JavaScript to improve reporting accuracy and data reliability.
- Worked directly with operators and engineers to gather user requirements, test applications, and refine tools for production use.
- Supported first-site deployment testing of the Squint platform by troubleshooting issues, documenting defects, and coordinating vendor improvements.
- Assisted with migration from Google Workspace to Microsoft 365 by rebuilding dashboards and redesigning reporting workflows.

Endangered Species International

Intern Leader

May 2022 – August 2022

- Led student teams in digital outreach and conservation awareness campaigns for rainforest protection.
- Coordinated weekly progress reviews and developed marketing materials that supported fundraising efforts.

Leadership and Projects

Alpha Sigma Kappa – Social Chair

August 2025 – May 2026

- Managed a large-scale social events budget exceeding \$1,000 and coordinated Greek Week activities.
- Organized events for groups of 200+ participants while ensuring safe, inclusive, and engaging environments across organizations.

COVID-19 Genome Sequencing and Alignment Project

August 2025 - December 2025

- Analyzed DNA sequence data using bioinformatics techniques to reconstruct and align genomic sequences against the SARS-CoV-2 reference genome.
- Performed sequence alignment and comparative analysis to identify similarities, differences, and potential mutations relative to the COVID-19 genome.
- Applied computational methods and data analysis tools to evaluate genomic variation and interpret biological significance.

Empower VR – Technology Lead

November 2024 – May 2025

- Led a cross-functional team of 10–12 students to develop a VR-based women's self-defense training application.
- Designed immersive learning features and coordinated closely with design and engineering teams to ensure cohesive functionality.
- Managed development milestones, addressed technical challenges, and kept the project on schedule.
- Helped prepare and submit a successful grant proposal that enabled the project to move forward.

Awards and Honors

Dean's Honor Roll – 2025 - Recognized for maintaining a technical GPA above a 3.66

Teacher's Assistant – 2024 - Provided targeted academic support to 10–20 students per week across multiple levels of mathematics.

State champion, HOSA Forensic Sciences - Won first prize in 2022 Illinois state championship of 1600 competitors.

Ambassadors Award for Exceptional Achievement in Sciences Member of National Honor Society - Maintained NHS requirements through high school including multiple volunteering and leadership opportunities in the community.